[INCH-POUND]

MIL-R-55182/13
4 October 1990

MILITARY SPECIFICATION

RESISTORS, FIXED, FILM
ESTABLISHED RELIABILITY
STYLE RNR77 1/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RNR77 1/, established reliability, film, fixed resistors.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-R-55182 - Resistors, Fixed, Film, Established Reliability, General Specification for.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5049.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-55182, and as specified herein.

1/ Third letter is variable, dependent upon lead material or capability.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

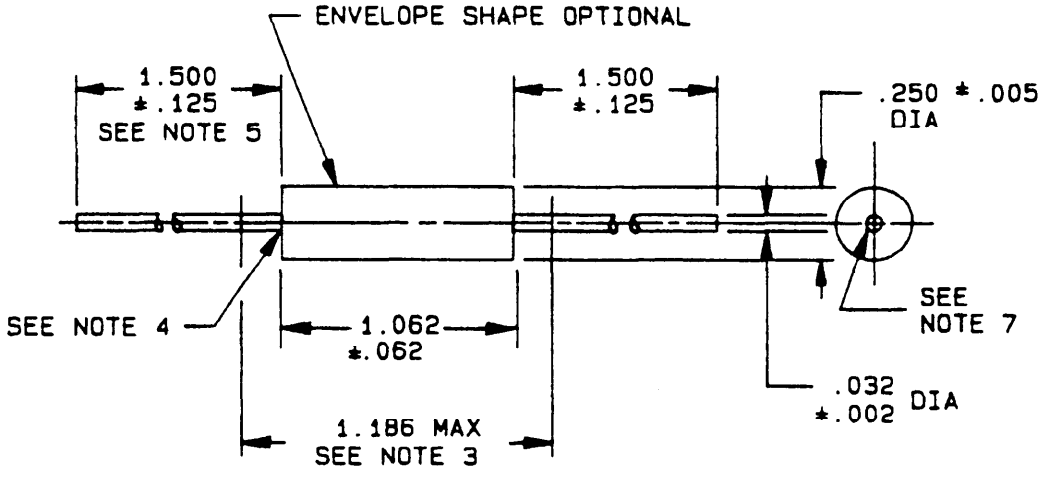
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3.2 Design, construction, and physical dimensions. The resistor shall be of the design, construction, and physical dimensions as specified on figure 1.

3.2.1 Characteristic. Style RNR77 is available in characteristics C, E, H, J, and K.

3.2.2 Terminal. Terminal type RNC is available in H, J, and K characteristics only. Terminal type RNR is available in characteristics C and E only.



Inches	mm
.002	0.05
.005	0.13
.031	0.79
.032	0.81
.062	1.57
.125	3.18
.240	6.10
1.062	26.97
1.186	30.12
1.500	38.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The maximum length is clean lead to clean lead.
4. The end of the body shall be that point at which the body diameter equals the nearest drill size larger than 250 percent of the nominal lead diamter.
5. Lead length for new design and tape and reel packaging shall be 1.00 +.625, -0.000 inch (25.4 +15.88, -0.000 mm).
6. Lead concentric tolerance is to be measured at the point of lead egress from the resistor body.
7. Lead to be concentric to the resistor body within .016 TIR.

FIGURE 1. Style RNR77 resistors.

3.3 Power rating. The power rating shall be 1.0 watt at 125°C (see 6.2).

3.4 Voltage rating. The continuous working voltage shall not exceed 750 volts.

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3.5 Resistance. Minimum and maximum resistance values shall be as follows:

Minimum resistance

Characteristic:

H, J, and K 2/	1.0 ohms
C and E	20.0 ohms

Maximum resistance

Characteristic:

C and E	20.0 megohms
H, J, and K	20.0 megohms

3.6 Voltage coefficient (applicable to resistors of 1,000 ohms and above). The voltage coefficient shall not exceed ±0.0005 percent per volt.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-55182.

4.2 Overload. The overload test duration shall be 1 hour with 2.25 x rated power overload; maximum voltage shall be 1,000 volts.

4.3 Terminal strength. The applied force shall be 5 pounds.

4.4 Dielectric withstanding voltage. The magnitude of test voltage shall be 900 volts for atmospheric pressure and 450 volts for barometric pressure.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-55182.

5.2 Marking. Packaging shall be marked with assigned part identification number, manufacturer's identification name, logo or CAGE code.

6. NOTES

6.1 Notes. In addition to the notes specified herein, the notes specified in MIL-R-55182 are applicable to this specification.

6.2 Power rating at 70°C. The power rating at 70°C for 2,000 hours life test has been established at 2.0 watts; however, it should be noted that the failure rate level is established at the single condition of 125°C, 1.0 watt and 10,000 hours life test duration. The user should consider the FR level as assigned only to the 125°C.

2/ For resistance tolerances D and F only. For resistance tolerance B, the minimum resistor value shall be 10.00 ohms.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR
Navy - AS, OS
Air Force - 17, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - CG, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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